

ONLINE RETAIL SALES PERFORMANCE AND CUSTOMER ANALYSIS

Executive Summary

This analysis examined online retail transaction dataset containing 541,909 transaction records to evaluate overall sales performance, product performance, customer contribution, geographical revenue distribution, and monthly sales trends. Following data cleaning and preparation, 530,104 valid records were retained for analysis.

The analysis generated approximately £10,666,685 in revenue, with the United Kingdom accounting for the majority of the revenue, approximately £9.03M. Product and customer analysis revealed differences between high-volume sales and high-revenue contributors, while monthly analysis showed substantial variation in performance across the period.

Based on these findings, recommendations were developed around geographical diversification, high-value customer retention, product performance optimization, and investigation of high and low performing sales periods.

Business questions:

This analysis is aimed at answering the following questions

- How is the overall performance of the business?
- What products and customers are driving the business?
- When and where is the business performing best?
- What countries generates the highest revenue?
- How does revenue change over time?

Dataset Overview

The dataset used for this project is an online retail transaction dataset containing records of customer purchases. Each row represents a transaction line containing information about the invoice, quantity purchased, transaction date, unit price, customer, and country. The original dataset contained 541,909 transaction records and 8 variables including; Invoice_No., Stock_Code, Description, Quantity, Invoice_Date, Unit_price, Customer_ID, and Country. After data cleaning and preparation, 530,104 valid records were retained for the analysis.

Dataset Summary

Metric	Value
Original records	541,909
Records retained after cleaning	530,104
Variables	8
Total revenue	£10,666,685

Total orders	19,960
Total customers	4,338
Total quantity sold	5.59 million
Average order value	£534.40

Data cleaning and preparation

The raw dataset was imported into power query for data cleaning and transformation. Data types were reviewed and appropriately assigned to ensure accurate calculations and analysis. The Invoice_No and stock_code fields were treated as text, quantity as a whole number, Invoice_Date as Date/Time field, Unit_Price as a decimal number, Customer_ID as a whole number and Country as text.

Cancelled transactions were identified using invoice numbers beginning with “C” and excluded from the analysis. Records with non-positive quantities and unit prices were also removed to ensure that revenue calculations represented valid sales transactions. A new revenue column was created by multiplying quantity purchased by unit price. A separate date-only field was also created from Invoice_Date to support monthly and time-based analysis in Power BI.

After the cleaning and transformation steps, the dataset contained 530,104 records, which were used for subsequent analysis and dashboard development.

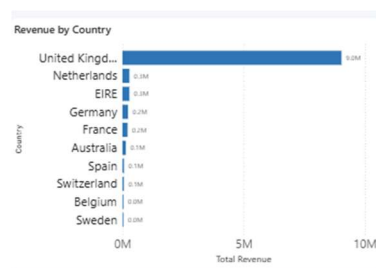
Tools and Technologies

The project was completed using a combination of data preparation, analysis, and visualization tools.

Tool	Purpose
Microsoft excel	Initial inspection and understanding of the dataset.
Power query	Data cleaning, transformation, and preparation.
Power BI	Data modeling, visualization, dashboard development, and analysis.
DAX	Creation of calculated measures and key performance indicators.

Key business insights

Insight 1 - Strong dependence on the UK market.

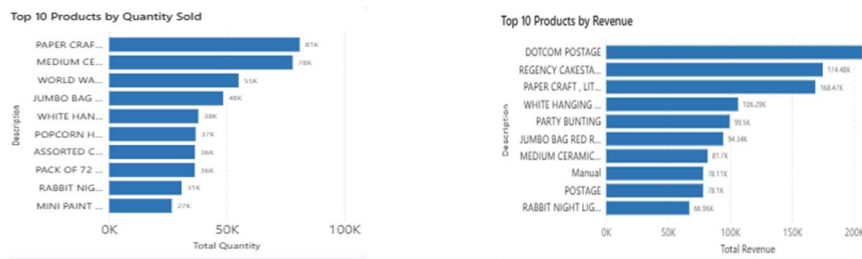


The United Kingdom generated £9.03M, making it by far the largest revenue market.

Business implication: The company has a strong established UK customer base, but this also creates geographical concentration risk.

Recommendation: Maintain UK customer retention while developing the Netherlands, Eire, Germany, France, Australia, and other markets.

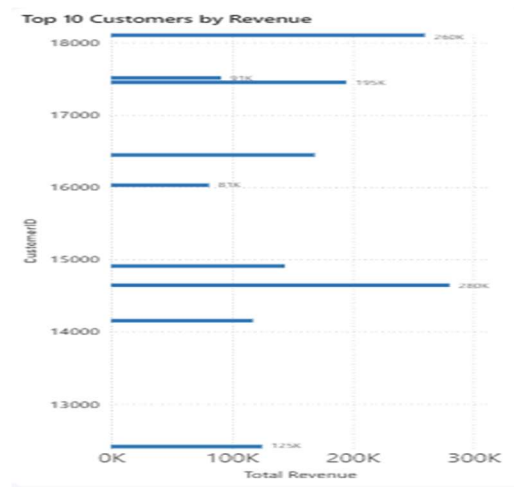
Insight 2 – High volume sales doesn't always mean highest revenue.



Paper Craft, little birdie sold 80,995 units, the highest quantity, while generating a total sum of £168,469.60 in revenue. Meanwhile, some products generated substantial revenue without appearing among the highest-volume products.

Business implication: Product volume and revenue contribution should be analyzed separately when making inventory and marketing decisions.

Insight 3 – Revenue is concentrated among a group of high value customers.



The highest revenue customer generated £280,206.02, while the 9 customers currently displayed in the top customer chart generated approximately £1.46M combined.

Business implication: Losing a high-value customer could have a noticeable effect on revenue.

Recommendation: Develop retention strategies for high-value customers, such as personalized offers, loyalty incentives, and targeted promotion.

Insight 4 – Large variation in monthly performance.



The month, November (2011), generated the highest revenue of £1,509,496.33, while February (2011), generated the lowest revenue of £523,631.89. November generated approximately 2.88x February's revenue.

Business implication: Understanding what factors that affected strongest months could help the business reproduce successful sales pattern.

Insight 5 – Non-merchandise items need separate consideration.

Some of the highest-revenue entries include Dotcom postage, postage, and manual. These aren't conventional merchandise products.

Business implication: Treating this as normal products could distort conclusions about which physical products are actually driving sales.

Recommendation: For future analysis, create a merchandise/non-merchandise classification so products performance can be evaluated more accurately.

Business Recommendation

1. Reduce dependence on the UK market: The business should maintain its strong UK customer base while increasing marketing and customer-acquisition efforts in other promising markets such as the Netherlands, Eire, Germany, France, and Australia.
Action: Identify the products and most popular in these markets and develop targeted promotions around them.
2. Protect and retain high-value customers: Introduce a customer-retention strategy focused on high-value customers.

Action: The retention strategy may include; personalized offers, loyalty rewards, early access to new products, targeted discounts, and re-engagement campaigns for customers whose purchases have declined.

3. Optimize inventory using both revenue and quantity: Inventory decisions should consider both quantity sold and revenue generated, rather than using sales volume alone.

Action: Classify products into;

- High volume / high revenue
- High volume / low revenue
- Low volume / high revenue
- Low volume / low revenue

These can help determine which products to promote, maintain, or reconsider.

4. Investigate peak and low sales period: Investigate the factors responsible for the strong November performance.

Action: Compare high-performing and low performing months by; product, country, number of orders, customer activity, and average order value. These could reveal seasonal patterns or successful promotional strategies that can be replicated.

5. Separate merchandise transactions from non-merchandise transactions: Create a separate classification for shipping/administrative items and physical merchandise.

Action: Use the merchandise-only analysis when making decisions about; product performance, inventory, product marketing, and product profitability. This will make future business conclusions more reliable.

The dashboard can be used as a recurring sales-monitoring tool, allowing management to monitor revenue, orders, customers, product, countries, and monthly performance through interactive filters.

Conclusion

The analysis demonstrates that the business generated substantial revenue but remains highly dependent on the UK market and a relatively small group of high-value customers. Product analysis also showed that sales volume and revenue contribution do not always follow the same pattern. Furthermore, significant differences in monthly revenue indicates opportunities to investigate seasonal and operational factor affecting performance. By strengthening customer retention, diversifying geographical markets, optimizing product decisions using multiple performance indicators, and investigating high-performing periods, the business can improve revenue stability and support more informed decision making.



Dashboard 1: Executive sales overview



Dashboard 2: Product and customer analysis



Dashboard 3: Sales performance trend